

Work Authorization: F-1 (CPT/OPT Eligible)

EDUCATION

Dartmouth College

B.A. Major in Computer Science, *Minor in Statistics (Completed)*

Hanover, NH

2023 – 2027

TECHNICAL SKILLS

Programming Languages:

Python, C/C++ , Java, Dart, JavaScript

Software Engineering:

Object-oriented design, asynchronous programming, REST APIs, state management

Mobile / Desktop:

Flutter (Web, Linux, Windows), responsive UI, platform-adaptive design

Backend / Cloud:

Firebase Authentication, Cloud Firestore, real-time data streams, JSON

Systems / Embedded:

Linux (CLI), SSH, Arduino, USB-Serial communication

Machine Learning:

PyTorch, scikit-learn, CNNs, model evaluation and experimentation

Developer Tools:

Git/GitHub, Vim/Neovim, SQL, HTML/CSS

EXPERIENCE

Dartmouth Center for Technology and Behavioral Health (Evergreen)

Software Engineer (Flutter / Cross-Platform)

Hanover, NH

2025 – Present

- Developed and shipped cross-platform Flutter features targeting Linux and Windows desktop environments.
- Designed and implemented a dual-mode *Developer vs User* interface, improving internal debugging efficiency while preserving clean end-user UX.
- Improved accessibility and UI correctness by resolving color-contrast issues in notification components.
- Collaborated with researchers on early-stage ML-driven feature design using ethically shared behavioral data.

AbleTo

Co-Founder & Lead Software Engineer

Remote (Peru / U.S.)

2024 – Present

- Architected and built the full software stack for *DotSense*, an autonomous Braille instruction system.
- Developed embedded C++ firmware for Arduino to control haptic actuators with precise 6-bit Braille pattern mapping.
- Built a Python (Tkinter) desktop application integrating speech recognition and text-to-speech for hands-free operation.
- Packaged and distributed a standalone Windows executable; validated end-to-end hardware–software integration.
- Recruited and led contributors from NYU, Georgia Tech, Texas A&M, Vanderbilt, UNI, and UNMSM.

Dartmouth College — EE Just Research Program

Machine Learning Research Intern

Hanover, NH

Summer 2024

- Developed end-to-end machine learning systems for medical prediction and diagnostic imaging using PyTorch, including data preprocessing, training, and validation pipelines.
- Evaluated neural network system performance (feedforward and CNNs) using quantitative metrics such as accuracy and confusion matrices in reproducible experimentation workflows.

SELECTED PROJECTS

Compiler Frontend (Lex/Yacc, C++)

2026

- Built a compiler frontend for a C-like language using Lex and Yacc, implementing tokenization, parsing, and AST construction.
- Designed grammar rules and semantic actions supporting expressions, control flow, and function definitions.
- Implemented structured error handling and symbol management consistent with real compiler pipelines.

DALI Social Platform — System Design Prototype (Flutter, Firebase)

2026

- Designed and implemented a prototype social platform demonstrating frontend–backend integration and real-time data modeling.
- Implemented email/password authentication, user discovery, profile pages, and Firestore-backed search.
- Identified state-management and UI consistency limitations, informing future architectural refactors.

COURSEWORK & TRAINING

Relevant Coursework:

Computer Architecture, Compilers, Discrete Mathematics, Probability, Linear Algebra, Statistics.

ML Training:

214+ hours (DataCamp) including supervised/unsupervised learning, CNNs, NLP, transformers, and PyTorch.

LEADERSHIP & HONORS

- **Beca Cometa Scholar (2023):** 1 of 2 students nationwide in Peru awarded a full undergraduate scholarship.
- **Joint Mathematics Meeting Presenter (2025):** Presented ML research on heart disease and brain tumor diagnosis.